

Final Exam
ECON1002, Econ Principles of Macro

August 8th, 2024

You have 115 minutes to complete this exam.

In the pursuit of the high ideals and rigorous standards of academic life, I commit myself to respect and uphold the Georgetown University Honor System:
To be honest in any academic endeavor, and To conduct myself honorably, as a responsible member of the Georgetown community, as we live and work together.
I agree that any code violations will result in a fail for this course.

Name: _____

Signature: _____

Question:	1	2	3	4	5	Total
Points:	18	14	34	34	15	115
Score:						

Instructions: Read all directions carefully and write your answers in the space provided. To receive full credit, you must show ***all*** of your work. Answers without supporting work will not be given credit. Make sure to sign the honor pledge statement above and to write your name. Make sure that you have all the exam pages.

1. (18 Points) Suppose the economy is initially in long-run equilibrium.

- 5 (a) Briefly explain why *SRAS* (short-run *AS*) and *LRAS* (long-run *AS*) curves have different shapes.
- 3 (b) Draw a graph representing the initial long-run equilibrium point A with *SRAS*, *LRAS*, and AD curves.
(*Hint: SRAS, LRAS, and AD curves intersect at the long-run equilibrium.*)
- 5 (c) Now assume a sudden negative cost shock hit the economy. What happens to the price level and aggregate output in the short-run equilibrium? Show the results graphically on the graph that you drew in part (a) by denoting the short-run equilibrium as point *B*.
- 5 (d) The government reacts by implementing an expansionary fiscal policy in response to the negative cost shock. What happens to the price level and aggregate output in the long-run equilibrium? Show the results graphically on the graph that you drew in part (a) by denoting the long-run equilibrium as point *C*.

2. (16 Points) Assume that the labor demand and labor supply curves in the United States are estimated as follows:

$$L_D = 175 - 4W$$

$$L_S = 16W - 30$$

where L is the quantity of labor demanded/supplied in millions of workers and W is the wage rate in dollars per hour.

- 5 (a) What is the market-clearing wage rate in this economy? Show the results graphically. Estimate the size of the labor force and the unemployment rate at the market-clearing wage.
- 4 (b) Does unemployment exist in the labor market at the market-clearing wage rate? Explain your answer by briefly mentioning the types of unemployment.
- 5 (c) Now suppose the minimum wage of \$12/hour is introduced to the labor market. Estimate the size of the labor force and the unemployment rate at the minimum wage rate of \$12/hour. Explain your answer by representing the impact of minimum wage on the graph that you drew in part (a).

3. (34 Points) Suppose Ukraine and Azerbaijan produce only two goods, wheat and cotton, using 100 acres of land for planting. You are given the following information:

Yield per Acre of Wheat of Cotton

	Ukraine	Azerbaijan
Wheat	6 bushels	1 bushel
Cotton	6 bales	3 bales

- 6 (a) Draw the production possibility frontier for each country in the absence of trade. Determine which country has the absolute advantage(s) in which goods.
- 5 (b) Assume that both countries consume equal amounts of wheat and cotton in the absence of trade. Denote the autarky consumption points as A on each country's PPF that you drew in part (a).
- 8 (c) Will Ukraine and Azerbaijan engage in free trade? Explain your answer by using the concept of comparative advantage.

- 5 (d) If a trade agreement is negotiated, at what rate would they agree to exchange? Find the range of terms of trade, cotton to wheat ratio (bales of cotton per 1 bushel of wheat).

- 10 (e) Now suppose the free trade is agreed with 2:1 terms of trade (cotton to wheat). Further, assume that after the agreement on free trade, Ukraine uses 75 acres of land for wheat and 25 acres of land for cotton, and Azerbaijan uses all 100 acres for cotton. Find the consumption points after the trade with agreed-to terms of trade for each country and denote these points as B on the graph that you drew in part (a).

4. (34 Points) Assume the following economy of a country (45 points):

- Consumption function: $C = 80 + 0.75Y_d$
- Investment: $I = 45$
- Government spending: $G = 100$
- Net taxes: $T = 20$
- Disposable income: $Y_d \equiv Y - T$
- Exports: $EX = 22$
- Imports: $IM = 0.05Y_d$

5

(a) Construct the (planned) aggregate expenditure function AE in this economy. Draw a graph showing AE and 45-degree line.

5

(b) Solve for the equilibrium income and denote the equilibrium point as E on the graph that you drew in part (a).

- 3 (c) When the economy is in equilibrium E , what is the government deficit? What is the current account balance?
- 8 (d) Now suppose that government spending G is increased to 115. What happens to equilibrium income? Explain by using a graphical illustration which represents new AE as AE' and new equilibrium as point A on the graph that you drew in part (a). What happens to imports?
- 3 (e) Find the government spending multiplier.
- 10 (f) Now, suppose the government targets a balanced current account. What must income be to ensure a current account balance of zero? By how much must we cut government spending to balance the current account?

15 5. (15 Points) Choose **ONE** question from the following six debatable macroeconomic issues and provide your opinion with compelling reason(s) to support your argument. When presenting your line of reasoning, make sure to mention a certain school of thoughts in macroeconomics that is closest to your thoughts.

- Should monetary and fiscal policymakers try to stabilize the economy?
- Should the government fight recessions with spending hikes rather than tax cuts?
- Should monetary policy be made by rule rather than by discretion?
- Should the central bank aim for zero inflation?
- Should the government balance its budget?
- Should the tax laws be reformed to encourage saving?

(Extra 15 Points) Show that the labor supply curve is backward bending when the income effect dominates the substitution effect.

This page is intentionally left blank to accommodate work that wouldn't fit elsewhere and/or scratch work.